

Dr. Parth Nayak

Data Scientist | AI Researcher | Astrophysicist



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 in [dr-parth-nayak-astro](#)
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EDUCATION

- 🎓 **Ph.D. in Astrophysics** (Dr. rer. nat.)
LMU Munich | Dec 2025
- 🎓 **B.Sc.+M.Sc. in Physics** (dual degree)
IISER Bhopal | Jul 2020

SKILLS

AI / Data Science

Deep machine learning (CNNs, MLPs, Transformers), simulation-based inference (SBI), evolutionary algorithms, Gaussian processes, Bayesian & Monte Carlo methods, density estimation

Programming

Python, LaTeX (proficient); HTML, CSS, Javascript (intermediate); C++, Java (basic)

Python packages

NumPy, SciPy, Pandas, Matplotlib, TensorFlow, Keras, Optuna, H5Py, AstroPy, HealPy, *among others*

OS / Shell

bash, Z shell (proficient), PowerShell (basic)

Other IT

High-performance computing (HPC), Git version control, LLM prompt engineering, Markdown, Mac/MS Office apps

LANGUAGES

English (full working proficiency); Gujarati, Hindi (native/bilingual); German (B1/limited working proficiency); French (beginner)

HIGHLIGHTS

- 5+ years of research experience in Data Science and AI algorithm development
- Led three and coauthored one research papers on novel AI applications in astrophysics
- Delivered 10+ technical talks at international conferences and research institutes
- Received 5+ academic awards and fellowships

RELEVANT EXPERIENCE

Doctoral Candidate: Astrophysics & Data Science LMU Munich | Oct 2021 - Dec 2025

- Developed a SBI framework “*LyaNNA*” [[par-nay/lyanna](#)] based on a novel 1D ResNet with Python/Keras from scratch that outperforms conventional methods by >100% in inference precision for complex, realistic, noisy spectroscopic data
- Developed “*SynTH*” [[par-nay/synth](#)] to model intergalactic absorption from massive cosmological hydrodynamic simulations and synthesize training data for *LyaNNA*
- Led two and coauthored one research papers on AI for inference in astrophysics [[Nayak+ 2024](#), [2025](#), [Chang+ 2025](#)]
- Conducted research at the Hightech Agenda Bayern Chair of Astrophysics, Cosmology & AI (ACAI) on AI applications in cosmology

Master's Thesis: Astrophysics & Data Science IISER Bhopal | Aug 2019 - Jul 2020

- Developed a GA framework “*GA-ILC*” to optimize the reconstruction of cosmic microwave background (CMB) from contaminated observations
 - ▶ A novel evolutionary algorithm [[par-nay/genepy](#)] to extract faint cosmological signals from high-noise environments (foreground removal)
- Published one research paper on evolutionary algorithms in astrophysics [[Nayak & Saha 2022](#)]

LEADERSHIP & SUPERVISION

- Co-supervisor of LMU Master's theses and labs Oct 2021 - Dec 2025
Formulated and supervised two LMU Master's theses, and co-supervised 5+ computational and instrumental lab courses
- Co-organizer of “Astronomy on Tap” (AoT) Munich since Oct 2025
Coordinating the monthly outreach event to engage the general public with accessible scientific talks and pub-quizzes
- Co-organizer of “Meet the Speaker” @LMU Observatory Oct 2023 - May 2025
Revitalized the weekly series by relocating it to the historic 200-year old telescope dome, a unique atmosphere for students for informal networking with international experts
- Coordinator & Senior Advisor of IISER Bhopal Astronomy Club (IBAC) 2016 - 2020
Played a foundational role as the club's second coordinator, establishing enduring traditions and growing the community into one of India's most popular student astronomy clubs
- Physics Department Representative Aug 2017 - Jul 2018
Unanimously elected to serve as the primary liaison between faculty and a class of 60+ physics major students

SELECTED AWARDS & FELLOWSHIPS

- DAAD Research Grant — Ph.D. scholarship Germany | Oct 2021 - Dec 2025
- Proficiency Medal in Physics — IISER Bhopal India | Jul 2020
- DAAD-WISE Scholarship Germany | May - Jul 2019
- INSPIRE Scholarship for Higher Education India | Aug 2015 - Jul 2020